



Kishoreganj University
3rd Year 1st Semester B.Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3103: Operating System (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.
Answer 05 out of 07 questions.

1. There are many computer systems (Batch to Handheld). Every computer system has its own functionalities, which is completely different or relates to each other's. User is one of the most vital parts of these systems. According to the system properties and user requirements we need different operating system to operate a computer.

- (a) Define operating system with example. Is it necessary for all systems? 3
 - (b) Which types of system is most user friendly among all systems? Explain briefly. 4
 - (c) Differentiate between a trap and an interrupt. 3
 - (d) Suppose you have four processes with length 48, 64, 120, and 28 bytes respectively. The memory available for user space is 300 bytes. Discuss a way by which you can run all the four processes parallelly in your system minimizing number of memory accesses. 4
- ✓ 2. (a) Explain the Convoy Effect of FCFS algorithm. Suggest one or more scheduling strategies that can reduce or eliminate the Convoy Effect, and briefly explain how they help. 2+2
- (b) Consider the following dataset: 3+1

Process	Arrival Time (s)	Burst Time (s)
P4	3	10
P2	0	5
P3	6	9
P5	7	12
P1	5	7

Draw the Gantt Chart and calculate the Response Time, Waiting Time and Turn Around Time of the Processes using Round Robin Algorithm by considering Time Quantum = 2s. In case the above scenario, if the context switch time is 3s, will there be any problem? $\neq$

- (c) A system uses preemptive priority scheduling, where a lower number means a higher priority. The CPU is allocated to the process with the highest priority, and preemption occurs whenever a new process with a higher priority arrives. The following processes arrive in the system: 3+3

Process	Arrival Time (ms)	Burst Time (ms)	Priority
P1	0	10	3
P2	2	4	1
P3	4	6	5
P4	6	2	4

Without Aging:

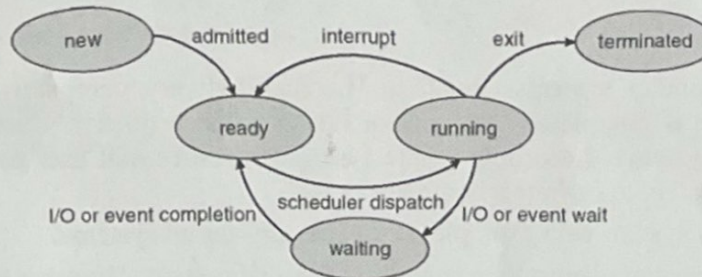
Draw the Gantt Chart showing process execution order based on preemptive priority scheduling algorithm. Also compute the average turnaround time, and average waiting time for the scenario.

With Aging:

Now assume the system implements aging to reduce the risk of starvation. Each process waiting in the ready queue will have its priority increased by 1 every 2ms of waiting. Redraw the Gantt chart considering the aging mechanism.

3. (a) Consider the following figure:

4



Assume there is a process named P1. Explain the life-cycle of the process P1 and how it transits from one state to another state according to the above figure.

- (b) Outline the procedures by which a parent process communicates with a child process. Illustrate the communication procedures with appropriate diagrams. 4
- (c) Consider the following semaphore code: 4+2

```

wait(S) {
while (S <= 0)
; // busy wait
S--;
}
signal(S){
S++;
}
  
```

Assume that initially, $S=2$ and there are three processes: P1, P2, and P3. Provide a step-by-step example showing how these processes use the semaphore to access the critical section. Determine whether this is a binary semaphore. Justify your answer.

4. (a) Define a safe state in operating systems. Under what circumstances should the Banker's Algorithm be preferred over the Resource Allocation Graph method for deadlock detection? 1+3
- (b) Consider the following snapshot of a system: 5+1

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P1	2	1	1	1	5	2	4	4	4	3	4	2
P2	3	2	2	1	4	2	6	2				
P3	2	1	1	2	2	3	1	7				
P4	1	4	3	1	2	5	3	4				

- i) Illustrate that whether the system is in safe state by demonstrating an order in which the process may arrive.
- ii) If a request from process P2 arrives for (0, 3, 0, 2) can the request be granted immediately?
- (c) Design a security defender software prototype to incorporate within the operating system of your computer to secure and protect your computer resources. 4

5. (a) Consider the following segment table:

4

Segment	Base	Length
0	2400	600
1	2300	50
2	1000	200
3	1270	480
4	1980	90

3, 1496
0, 400

Draw the flowchart for converting Logical Address to Physical Address and identify the location of the physical addresses in the main memory for the following logical addresses.

- (b) Differentiate between internal fragmentation and external fragmentation with proper examples. 4
- (c) Given memory partitions of 220K, 55K, 20K, 120K, 75K, and 35K (in order), how would Best Fit and Worst Fit algorithms place each of the processes P1 to P10 given in the table (in order). 6

Process	Size	Turnaround
P1	50K	2
P2	20K	3
P3	35K	2
P4	110K	3
P5	90K	4
P6	200K	1
P7	5K	2
P8	55K	1
P9	170K	3
P10	86K	1

Memory Block	Size
Block 1	220K
Block 2	55K
Block 3	20K
Block 4	120K
Block 5	75K
Block 6	35K

6. (a) Explain Demand Paging with proper diagram. Identify the tasks of dirty bits and protection bits of the page table. 3+1
- (b) Consider the following reference string for a page replacement system: 4
- Reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 ~ 4 5 2 3
- Using FIFO (First-In-First-Out) page replacement algorithm, calculate the number of page faults:
- When the number of frames is 3
 - When the number of frames is 4
- Based on the above results, explain Belady's anomaly.
- (c) Consider the following page references: 6

5, 6, 4, 5, 7, 9, 2, 4, 3, 7, 4, 6, 4, 9, 4, 6, 7, 8, 3, 5, 2, 1, 3, 4, 6, 2, 4, 3

Assuming demand paging with four frames and the frames are given below:

4
5
6

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Apply LRU (Least Recently Used), and Optimal Page Replacement algorithms, and identify the algorithm that shows the best performance in this context.

7. (a) A disk has 50 blocks (numbered 0 to 49), and each block is 1 KB in size. The disk currently has the following files stored using contiguous allocation: 3+1

File	Starting Block	Length (in blocks)
A	0	10
B	12	5
C	20	8

Draw a schematic diagram of the disk showing the locations of files A, B, C, and the free blocks. A new file D of size 6 KB needs to be stored. Can it be stored contiguously? If yes, suggest the starting block; if not, explain why.

- (b) Describe acyclic-graph directory structure in a file system. Analyze the methods used by the operating system to prevent cycles when linking directories. 2+2
- (c) Suppose a disk drive has 2000 tracks (0-1999) and the disk queue having I/O requests in the following order as follows: 3+3

100, 250, 1500, 750, 950, 200, 1450, 850, 150, 1850, 550, 1100, 1720, 90, 310, 280, 350, 480, 1375, 1530, 1780, 1890, 1410, 270, 170

Current position of the Read-Write head is 400 and assume that previous request was at 345. Starting from the current head position, identify the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms:

- SSTF
- SCAN